SG No. 89 D_4^1 $P422$ [tetragonal]

* plus set: $+[0, 0, 0]$
* Wyckoff site: `1a`, site symmetry: `422`

Table 1: Wyckoff bond: `1a@1a`, bond symmetry: `422`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[0, 0, 0]$	`[1,2,3,4,-5,-6,-7,-8]`

Table 2: Wyckoff bond: `2b@1a`, bond symmetry: `222.`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[0, 0, 0]$	`[1,-2,-5,6]`
2	$[0, X, 0]$	$[0, 0, 0]$	`[3,-4,7,-8]`

Table 3: Wyckoff bond: `2c@1a`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[X, X, 0]$	$[0, 0, 0]$	`[1,-2,7,-8]`
2	$[-X, X, 0]$	$[0, 0, 0]$	`[3,-4,5,-6]`

Table 4: Wyckoff bond: `4d@1a`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$[0, 0, 0]$	`[1,-5]`
2	$[-X, 0, Z]$	$[0, 0, 0]$	`[2,-6]`
3	$[0, X, Z]$	$[0, 0, 0]$	`[3,-8]`
4	$[0, -X, Z]$	$[0, 0, 0]$	`[4,-7]`

Table 5: Wyckoff bond: `4e@1a`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, X, Z]$	$[0, 0, 0]$	`[1,-8]`
2	$[-X, -X, Z]$	$[0, 0, 0]$	`[2,-7]`
3	$[-X, X, Z]$	$[0, 0, 0]$	`[3,-6]`
4	$[X, -X, Z]$	$[0, 0, 0]$	`[4,-5]`

Table 6: Wyckoff bond: `4f@1a`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[0, 0, 0]$	`[1,-2]`
2	$[-Y, X, 0]$	$[0, 0, 0]$	`[3,-4]`
3	$[-X, Y, 0]$	$[0, 0, 0]$	`[5,-6]`

4	$[Y, X, 0]$	$[0, 0, 0]$	`[7,-8]`

Table 7: Wyckoff bond: `8g@1a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, 0]$	`[1]`
2	$[-X, -Y, Z]$	$[0, 0, 0]$	`[2]`
3	$[-Y, X, Z]$	$[0, 0, 0]$	`[3]`
4	$[Y, -X, Z]$	$[0, 0, 0]$	`[4]`
5	$[-X, Y, -Z]$	$[0, 0, 0]$	`[5]`
6	$[X, -Y, -Z]$	$[0, 0, 0]$	`[6]`
7	$[Y, X, -Z]$	$[0, 0, 0]$	`[7]`
8	$[-Y, -X, -Z]$	$[0, 0, 0]$	`[8]`

* Wyckoff site: `1b`, site symmetry: `422`

Table 8: Wyckoff bond: `1a@1b`, bond symmetry: `422`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,2,3,4,-5,-6,-7,-8]`

Table 9: Wyckoff bond: `2b@1b`, bond symmetry: `222.`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,-2,-5,6]`
2	$[0, X, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[3,-4,7,-8]`

Table 10: Wyckoff bond: `2c@1b`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[X, X, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,-2,7,-8]`
2	$[-X, X, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[3,-4,5,-6]`

Table 11: Wyckoff bond: `4d@1b`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,-5]`
2	$[-X, 0, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[2,-6]`
3	$[0, X, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[3,-8]`
4	$[0, -X, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[4,-7]`

Table 12: Wyckoff bond: `4e@1b`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, X, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,-8]`
2	$[-X, -X, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[2,-7]`
3	$[-X, X, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[3,-6]`
4	$[X, -X, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[4,-5]`

Table 13: Wyckoff bond: `4f@1b`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,-2]`
2	$[-Y, X, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[3,-4]`
3	$[-X, Y, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[5,-6]`
4	$[Y, X, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[7,-8]`

Table 14: Wyckoff bond: `8g@1b`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[8]`

* Wyckoff site: `1c`, site symmetry: `422`

Table 15: Wyckoff bond: `1a@1c`, bond symmetry: `422`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,2,3,4,-5,-6,-7,-8]`

Table 16: Wyckoff bond: `2b@1c`, bond symmetry: `222.`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,-2,-5,6]`
2	$[0, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,-4,7,-8]`

Table 17: Wyckoff bond: `2c@1c`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[X, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,-2,7,-8]`

continued ...

Table 17

No.	vector	center	mapping
2	$[-X, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,-4,5,-6]`

Table 18: Wyckoff bond: `4d@1c`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,-5]`
2	$[-X, 0, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[2,-6]`
3	$[0, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,-8]`
4	$[0, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[4,-7]`

Table 19: Wyckoff bond: `4e@1c`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,-8]`
2	$[-X, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[2,-7]`
3	$[-X, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,-6]`
4	$[X, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[4,-5]`

Table 20: Wyckoff bond: `4f@1c`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,-2]`
2	$[-Y, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,-4]`
3	$[-X, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[5,-6]`
4	$[Y, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[7,-8]`

Table 21: Wyckoff bond: `8g@1c`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[8]`

* Wyckoff site: `1d`, site symmetry: `422`

Table 22: Wyckoff bond: `1a@1d`, bond symmetry: `422`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,2,3,4,-5,-6,-7,-8]`

Table 23: Wyckoff bond: `2b@1d`, bond symmetry: `222.`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-2,-5,6]`
2	$[0, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[3,-4,7,-8]`

Table 24: Wyckoff bond: `2c@1d`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[X, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-2,7,-8]`
2	$[-X, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[3,-4,5,-6]`

Table 25: Wyckoff bond: `4d@1d`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-5]`
2	$[-X, 0, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[2,-6]`
3	$[0, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[3,-8]`
4	$[0, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[4,-7]`

Table 26: Wyckoff bond: `4e@1d`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-8]`
2	$[-X, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[2,-7]`
3	$[-X, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[3,-6]`
4	$[X, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[4,-5]`

Table 27: Wyckoff bond: `4f@1d`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-2]`
2	$[-Y, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[3,-4]`
3	$[-X, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[5,-6]`
4	$[Y, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[7,-8]`

Table 28: Wyckoff bond: `8g@1d`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[8]`

* Wyckoff site: `2e`, site symmetry: `222.`

Table 29: Wyckoff bond: `2a@2e`, bond symmetry: `222.`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1,-2,-5,6]`
2	$[0, X, 0]$	$\left[0, \frac{1}{2}, 0\right]$	`[3,-4,7,-8]`

Table 30: Wyckoff bond: `2b@2e`, bond symmetry: `222.`

No.	vector	center	mapping
1	$[0, X, 0]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1,-2,5,-6]`
2	$[-X, 0, 0]$	$\left[0, \frac{1}{2}, 0\right]$	`[3,-4,-7,8]`

Table 31: Wyckoff bond: `2c@2e`, bond symmetry: `222.`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1,2,-5,-6]`
2	$[0, 0, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[3,4,-7,-8]`

Table 32: Wyckoff bond: `4d@2e`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1,-5]`
2	$[-X, 0, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[2,-6]`
3	$[0, X, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[3,-8]`
4	$[0, -X, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[4,-7]`

Table 33: Wyckoff bond: `4e@2e`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[0, X, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1,-6]`
2	$[0, -X, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[2,-5]`

continued ...

Table 33

No.	vector	center	mapping
3	$[-X, 0, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[3,-7]`
4	$[X, 0, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[4,-8]`

Table 34: Wyckoff bond: `4f@2e`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1,-2]`
2	$[-Y, X, 0]$	$\left[0, \frac{1}{2}, 0\right]$	`[3,-4]`
3	$[-X, Y, 0]$	$\left[\frac{1}{2}, 0, 0\right]$	`[5,-6]`
4	$[Y, X, 0]$	$\left[0, \frac{1}{2}, 0\right]$	`[7,-8]`

Table 35: Wyckoff bond: `8g@2e`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[8]`

* Wyckoff site: `2f`, site symmetry: `222.`

Table 36: Wyckoff bond: `2a@2f`, bond symmetry: `222.`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[1,-2,-5,6]`
2	$[0, X, 0]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[3,-4,7,-8]`

Table 37: Wyckoff bond: `2b@2f`, bond symmetry: `222.`

No.	vector	center	mapping
1	$[0, X, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[1,-2,5,-6]`
2	$[-X, 0, 0]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[3,-4,-7,8]`

Table 38: Wyckoff bond: `2c@2f`, bond symmetry: `222.`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[1,2,-5,-6]`
2	$[0, 0, Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[3,4,-7,-8]`

Table 39: Wyckoff bond: `4d@2f`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[1,-5]`
2	$[-X, 0, Z]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[2,-6]`
3	$[0, X, Z]$	$[0, \frac{1}{2}, \frac{1}{2}]$	`[3,-8]`
4	$[0, -X, Z]$	$[0, \frac{1}{2}, \frac{1}{2}]$	`[4,-7]`

Table 40: Wyckoff bond: `4e@2f`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[0, X, Z]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[1,-6]`
2	$[0, -X, Z]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[2,-5]`
3	$[-X, 0, Z]$	$[0, \frac{1}{2}, \frac{1}{2}]$	`[3,-7]`
4	$[X, 0, Z]$	$[0, \frac{1}{2}, \frac{1}{2}]$	`[4,-8]`

Table 41: Wyckoff bond: `4f@2f`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[1,-2]`
2	$[-Y, X, 0]$	$[0, \frac{1}{2}, \frac{1}{2}]$	`[3,-4]`
3	$[-X, Y, 0]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[5,-6]`
4	$[Y, X, 0]$	$[0, \frac{1}{2}, \frac{1}{2}]$	`[7,-8]`

Table 42: Wyckoff bond: `8g@2f`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[1]`
2	$[-X, -Y, Z]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[2]`
3	$[-Y, X, Z]$	$[0, \frac{1}{2}, \frac{1}{2}]$	`[3]`
4	$[Y, -X, Z]$	$[0, \frac{1}{2}, \frac{1}{2}]$	`[4]`
5	$[-X, Y, -Z]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[5]`
6	$[X, -Y, -Z]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[6]`
7	$[Y, X, -Z]$	$[0, \frac{1}{2}, \frac{1}{2}]$	`[7]`
8	$[-Y, -X, -Z]$	$[0, \frac{1}{2}, \frac{1}{2}]$	`[8]`

* Wyckoff site: `2g`, site symmetry: `4..`

Table 43: Wyckoff bond: `2a@2g`, bond symmetry: `4..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[0, 0, z]$	`[1,2,3,4]`
2	$[0, 0, -Z]$	$[0, 0, -z]$	`[5,6,7,8]`

Table 44: Wyckoff bond: `4b@2g`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[0, 0, z]$	`[1,-2]`
2	$[-Y, X, 0]$	$[0, 0, z]$	`[3,-4]`
3	$[-X, Y, 0]$	$[0, 0, -z]$	`[5,-6]`
4	$[Y, X, 0]$	$[0, 0, -z]$	`[7,-8]`

Table 45: Wyckoff bond: `8c@2g`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, z]$	`[1]`
2	$[-X, -Y, Z]$	$[0, 0, z]$	`[2]`
3	$[-Y, X, Z]$	$[0, 0, z]$	`[3]`
4	$[Y, -X, Z]$	$[0, 0, z]$	`[4]`
5	$[-X, Y, -Z]$	$[0, 0, -z]$	`[5]`
6	$[X, -Y, -Z]$	$[0, 0, -z]$	`[6]`
7	$[Y, X, -Z]$	$[0, 0, -z]$	`[7]`
8	$[-Y, -X, -Z]$	$[0, 0, -z]$	`[8]`

* Wyckoff site: `2h`, site symmetry: `4..`

Table 46: Wyckoff bond: `2a@2h`, bond symmetry: `4..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[1,2,3,4]`
2	$[0, 0, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[5,6,7,8]`

Table 47: Wyckoff bond: `4b@2h`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[1,-2]`
2	$[-Y, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[3,-4]`
3	$[-X, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[5,-6]`
4	$[Y, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[7,-8]`

Table 48: Wyckoff bond: `8c@2h`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[7]`

continued ...

Table 48

No.	vector	center	mapping
8	$[-Y, -X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[8]`

* Wyckoff site: `4i`, site symmetry: `2..`

Table 49: Wyckoff bond: `4a@4i`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[0, \frac{1}{2}, z\right]$	`[1,-2]`
2	$[-Y, X, 0]$	$\left[\frac{1}{2}, 0, z\right]$	`[3,-4]`
3	$[-X, Y, 0]$	$\left[0, \frac{1}{2}, -z\right]$	`[5,-6]`
4	$[Y, X, 0]$	$\left[\frac{1}{2}, 0, -z\right]$	`[7,-8]`

Table 50: Wyckoff bond: `4b@4i`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[1,2]`
2	$[0, 0, Z]$	$\left[\frac{1}{2}, 0, z\right]$	`[3,4]`
3	$[0, 0, -Z]$	$\left[0, \frac{1}{2}, -z\right]$	`[5,6]`
4	$[0, 0, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	`[7,8]`

Table 51: Wyckoff bond: `8c@4i`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[\frac{1}{2}, 0, z\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[\frac{1}{2}, 0, z\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[0, \frac{1}{2}, -z\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[0, \frac{1}{2}, -z\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	`[8]`

* Wyckoff site: `4j`, site symmetry: `..2`

Table 52: Wyckoff bond: `4a@4j`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, -X, Z]$	$[x, x, 0]$	`[1,-7]`
2	$[-X, X, Z]$	$[-x, -x, 0]$	`[2,-8]`
3	$[X, X, Z]$	$[-x, x, 0]$	`[3,-5]`
4	$[-X, -X, Z]$	$[x, -x, 0]$	`[4,-6]`

Table 53: Wyckoff bond: `4b@4j`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, X, 0]$	$[x, x, 0]$	`[1,7]`
2	$[-X, -X, 0]$	$[-x, -x, 0]$	`[2,8]`
3	$[-X, X, 0]$	$[-x, x, 0]$	`[3,5]`
4	$[X, -X, 0]$	$[x, -x, 0]$	`[4,6]`

Table 54: Wyckoff bond: `8c@4j`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, x, 0]$	`[1]`
2	$[-X, -Y, Z]$	$[-x, -x, 0]$	`[2]`
3	$[-Y, X, Z]$	$[-x, x, 0]$	`[3]`
4	$[Y, -X, Z]$	$[x, -x, 0]$	`[4]`
5	$[-X, Y, -Z]$	$[-x, x, 0]$	`[5]`
6	$[X, -Y, -Z]$	$[x, -x, 0]$	`[6]`
7	$[Y, X, -Z]$	$[x, x, 0]$	`[7]`
8	$[-Y, -X, -Z]$	$[-x, -x, 0]$	`[8]`

* Wyckoff site: `4k`, site symmetry: `..2`

Table 55: Wyckoff bond: `4a@4k`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, -X, Z]$	$\left[x, x, \frac{1}{2}\right]$	`[1,-7]`
2	$[-X, X, Z]$	$\left[-x, -x, \frac{1}{2}\right]$	`[2,-8]`
3	$[X, X, Z]$	$\left[-x, x, \frac{1}{2}\right]$	`[3,-5]`
4	$[-X, -X, Z]$	$\left[x, -x, \frac{1}{2}\right]$	`[4,-6]`

Table 56: Wyckoff bond: `4b@4k`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, X, 0]$	$\left[x, x, \frac{1}{2}\right]$	`[1,7]`
2	$[-X, -X, 0]$	$\left[-x, -x, \frac{1}{2}\right]$	`[2,8]`
3	$[-X, X, 0]$	$\left[-x, x, \frac{1}{2}\right]$	`[3,5]`
4	$[X, -X, 0]$	$\left[x, -x, \frac{1}{2}\right]$	`[4,6]`

Table 57: Wyckoff bond: `8c@4k`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, x, \frac{1}{2}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[-x, -x, \frac{1}{2}\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[-x, x, \frac{1}{2}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[x, -x, \frac{1}{2}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[-x, x, \frac{1}{2}\right]$	`[5]`

continued ...

Table 57

No.	vector	center	mapping
6	$[X, -Y, -Z]$	$\left[x, -x, \frac{1}{2}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[x, x, \frac{1}{2}\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[-x, -x, \frac{1}{2}\right]$	`[8]`

* Wyckoff site: `4l`, site symmetry: `.2.`

Table 58: Wyckoff bond: `4a@4l`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[0, X, Z]$	$[x, 0, 0]$	`[1,-6]`
2	$[0, -X, Z]$	$[-x, 0, 0]$	`[2,-5]`
3	$[-X, 0, Z]$	$[0, x, 0]$	`[3,-7]`
4	$[X, 0, Z]$	$[0, -x, 0]$	`[4,-8]`

Table 59: Wyckoff bond: `4b@4l`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[x, 0, 0]$	`[1,6]`
2	$[-X, 0, 0]$	$[-x, 0, 0]$	`[2,5]`
3	$[0, X, 0]$	$[0, x, 0]$	`[3,7]`
4	$[0, -X, 0]$	$[0, -x, 0]$	`[4,8]`

Table 60: Wyckoff bond: `8c@4l`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, 0, 0]$	`[1]`
2	$[-X, -Y, Z]$	$[-x, 0, 0]$	`[2]`
3	$[-Y, X, Z]$	$[0, x, 0]$	`[3]`
4	$[Y, -X, Z]$	$[0, -x, 0]$	`[4]`
5	$[-X, Y, -Z]$	$[-x, 0, 0]$	`[5]`
6	$[X, -Y, -Z]$	$[x, 0, 0]$	`[6]`
7	$[Y, X, -Z]$	$[0, x, 0]$	`[7]`
8	$[-Y, -X, -Z]$	$[0, -x, 0]$	`[8]`

* Wyckoff site: `4m`, site symmetry: `.2.`

Table 61: Wyckoff bond: `4a@4m`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[0, X, Z]$	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-6]`
2	$[0, -X, Z]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	`[2,-5]`
3	$[-X, 0, Z]$	$\left[\frac{1}{2}, x, \frac{1}{2}\right]$	`[3,-7]`
4	$[X, 0, Z]$	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	`[4,-8]`

Table 62: Wyckoff bond: `4b@4m`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	`[1,6]`
2	$[-X, 0, 0]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	`[2,5]`
3	$[0, X, 0]$	$\left[\frac{1}{2}, x, \frac{1}{2}\right]$	`[3,7]`
4	$[0, -X, 0]$	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	`[4,8]`

Table 63: Wyckoff bond: `8c@4m`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[\frac{1}{2}, x, \frac{1}{2}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[\frac{1}{2}, x, \frac{1}{2}\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	`[8]`

* Wyckoff site: `4n`, site symmetry: `.2.`

Table 64: Wyckoff bond: `4a@4n`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[0, X, Z]$	$\left[x, 0, \frac{1}{2}\right]$	`[1,-6]`
2	$[0, -X, Z]$	$\left[-x, 0, \frac{1}{2}\right]$	`[2,-5]`
3	$[-X, 0, Z]$	$\left[0, x, \frac{1}{2}\right]$	`[3,-7]`
4	$[X, 0, Z]$	$\left[0, -x, \frac{1}{2}\right]$	`[4,-8]`

Table 65: Wyckoff bond: `4b@4n`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[x, 0, \frac{1}{2}\right]$	`[1,6]`
2	$[-X, 0, 0]$	$\left[-x, 0, \frac{1}{2}\right]$	`[2,5]`
3	$[0, X, 0]$	$\left[0, x, \frac{1}{2}\right]$	`[3,7]`
4	$[0, -X, 0]$	$\left[0, -x, \frac{1}{2}\right]$	`[4,8]`

Table 66: Wyckoff bond: `8c@4n`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, 0, \frac{1}{2}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[-x, 0, \frac{1}{2}\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[0, x, \frac{1}{2}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[0, -x, \frac{1}{2}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[-x, 0, \frac{1}{2}\right]$	`[5]`

continued ...

Table 66

No.	vector	center	mapping
6	$[X, -Y, -Z]$	$\left[x, 0, \frac{1}{2}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[0, x, \frac{1}{2}\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[0, -x, \frac{1}{2}\right]$	`[8]`

* Wyckoff site: `4o`, site symmetry: `.2.`

Table 67: Wyckoff bond: `4a@4o`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[0, X, Z]$	$\left[x, \frac{1}{2}, 0\right]$	`[1,-6]`
2	$[0, -X, Z]$	$\left[-x, \frac{1}{2}, 0\right]$	`[2,-5]`
3	$[-X, 0, Z]$	$\left[\frac{1}{2}, x, 0\right]$	`[3,-7]`
4	$[X, 0, Z]$	$\left[\frac{1}{2}, -x, 0\right]$	`[4,-8]`

Table 68: Wyckoff bond: `4b@4o`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[x, \frac{1}{2}, 0\right]$	`[1,6]`
2	$[-X, 0, 0]$	$\left[-x, \frac{1}{2}, 0\right]$	`[2,5]`
3	$[0, X, 0]$	$\left[\frac{1}{2}, x, 0\right]$	`[3,7]`
4	$[0, -X, 0]$	$\left[\frac{1}{2}, -x, 0\right]$	`[4,8]`

Table 69: Wyckoff bond: `8c@4o`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, \frac{1}{2}, 0\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[-x, \frac{1}{2}, 0\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[\frac{1}{2}, x, 0\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[\frac{1}{2}, -x, 0\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[-x, \frac{1}{2}, 0\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[x, \frac{1}{2}, 0\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[\frac{1}{2}, x, 0\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[\frac{1}{2}, -x, 0\right]$	`[8]`

* Wyckoff site: `8p`, site symmetry: `1`

Table 70: Wyckoff bond: `8a@8p`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, z]$	`[1]`
2	$[-X, -Y, Z]$	$[-x, -y, z]$	`[2]`
3	$[-Y, X, Z]$	$[-y, x, z]$	`[3]`
4	$[Y, -X, Z]$	$[y, -x, z]$	`[4]`
5	$[-X, Y, -Z]$	$[-x, y, -z]$	`[5]`

continued ...

Table 70

No.	vector	center	mapping
6	$[X, -Y, -Z]$	$[x, -y, -z]$	`[6]`
7	$[Y, X, -Z]$	$[y, x, -z]$	`[7]`
8	$[-Y, -X, -Z]$	$[-y, -x, -z]$	`[8]`